NUR 774: Ear Pictures Explained

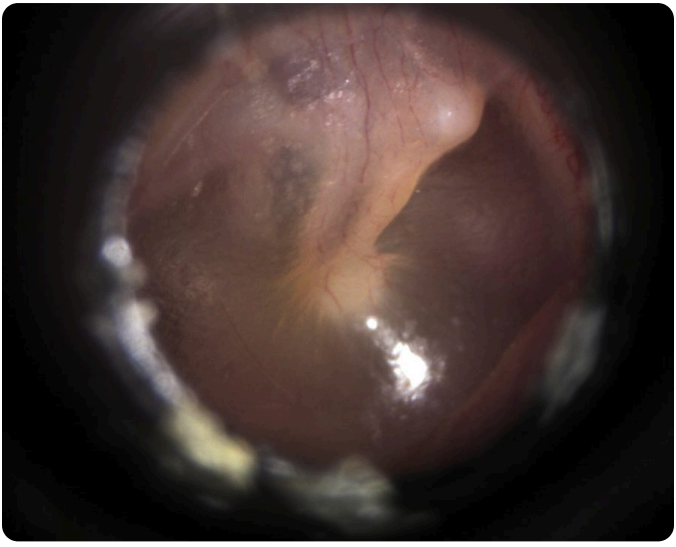
Every picture from the Normal and Abnormal Middle Ear Anatomy slides: what it is, what you're looking at, a mnemonic, and practice questions with answers and rationales.

How to use: look at each picture and name it first, then swipe for the answer. Questions follow each picture, with the answer on the next page.

Quick ID cheat sheet

- **Slide 2:** Normal tympanic membrane
- **Slide 3:** Normal TM with labeled landmarks
- **Slide 4:** Normal left ear (incus visible)
- **Slide 5:** Red TM from crying or fever (NOT AOM)
- **Slide 6:** Air-fluid level with bubbles (OME)
- **Slide 7:** Amber fluid with air-fluid level (OME)
- **Slide 8:** Acute otitis media: moderate bulging
- **Slide 9:** AOM with bulging and TM hemorrhage
- **Slide 10:** Perforated tympanic membrane
- **Slide 11:** Bullous myringitis
- **Slide 12:** Tympanosclerosis
- **Slide 13:** Attic cholesteatoma
- **Slide 14:** Acute otitis externa with pinna cellulitis/perichondritis
- **Slide 15:** Bacterial acute otitis externa (inside the canal)
- **Slide 16:** Otitis externa (swollen canal opening)
- **Slide 17:** Otitis externa (narrowed canal on otoscopy)

Slide 2: what is this?



Name it before you swipe →

Slide 2

Normal tympanic membrane

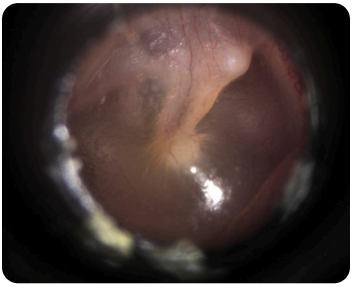


Photo: Hawke, M. Hawke Library (course slides).

What it is: A healthy eardrum (TM). This is what every other picture is compared against.

What you see:

- Pearly gray and see-through (translucent): you can partly see the middle ear behind it
- The handle of the malleus (the little bone) is easy to see
- A bright cone of light reflects off the lower front part
- No fluid, no bubbles, no bulging, no redness

Slide 2: remember it

Normal tympanic membrane

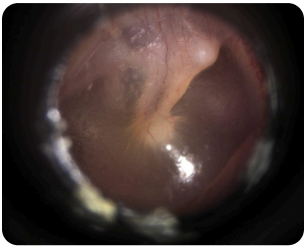
> 🧠 **Normal = "PEARL": Pearly gray, Easy-to-see landmarks, Air (no fluid) behind it, Reflex (cone of light), Lies flat (no bulging) and moves with a puff of air.**

Exam tips:

- The cone of light points to about 5 o'clock in the RIGHT ear and 7 o'clock in the LEFT ear ("Right 5, Left 7").
- A normal TM moves with pneumatic otoscopy. A TM that doesn't move means fluid (OME or AOM).

Question 1 (slide 2)

SELECT ALL THAT APPLY



Which findings describe a NORMAL tympanic membrane?

A. Cone of light present

B. Bulging with loss of landmarks

C. Moves with pneumatic otoscopy

D. Amber fluid behind the drum

E. Pearly gray and translucent

F. Visible handle of the malleus

Swipe for answer →

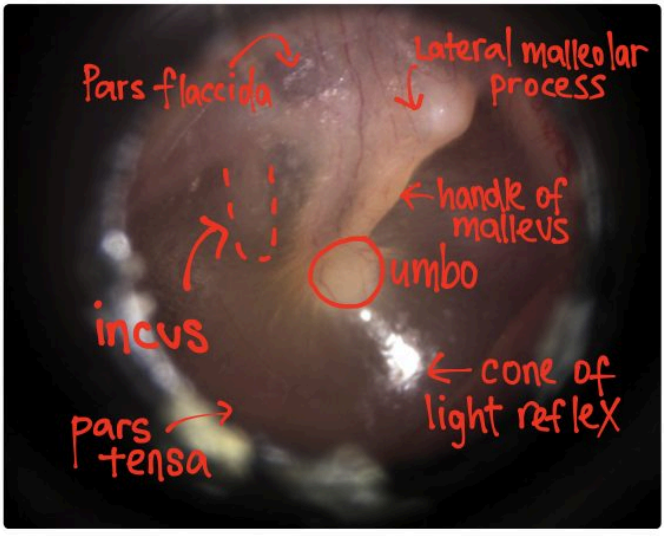
Answer 1

A, C, E, F

- **Cone of light present**
- **Moves with pneumatic otoscopy**
- **Pearly gray and translucent**
- **Visible handle of the malleus**

Why: Normal = PEARL. Amber fluid = OME. Bulging with lost landmarks = AOM.

Slide 3: what is this?



Name it before you swipe →

Slide 3

Normal TM with labeled landmarks

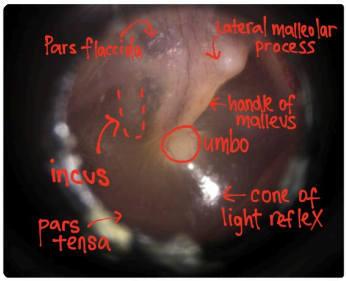


Photo: Hawke, M. Hawke Library (course slides).

What it is: The same normal eardrum, with the parts labeled. Exams love asking you to name these.

What you see:

- Pars flaccida: the small, floppy top part of the eardrum
- Lateral malleolar process: the small bump at the top of the malleus
- Handle of the malleus: the bone running down the middle
- Umbo: the center point where the malleus ends (the "belly button")
- Cone of light: the bright triangle below the umbo
- Pars tensa: the large, tight main part of the eardrum
- Incus: a second little bone you can sometimes see behind the drum

Normal TM with labeled landmarks

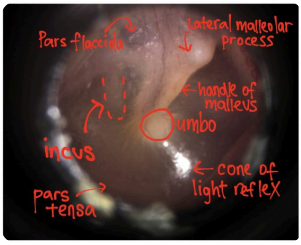
> 🧠 **Read it top to bottom like a lollipop: "Floppy top, Knob, Handle, Belly button, Light." Pars Flaccida (Floppy) → lateral process (Knob) → Handle of malleus → Umbo (belly button) → cone of Light, all sitting on the Pars Tensa (Tight).**

Exam tips:

- When AOM makes the TM bulge, these landmarks DISAPPEAR. "Loss of landmarks" is a key AOM phrase.
- Cholesteatoma often starts in the pars flaccida (see slide 13).

Question 2 (slide 3)

SINGLE ANSWER



The nurse sees the central point of the eardrum where the handle of the malleus ends. This landmark is the:

A. Umbo

B. Incus

C. Cone of light

D. Pars flaccida

Swipe for answer →

Answer 2

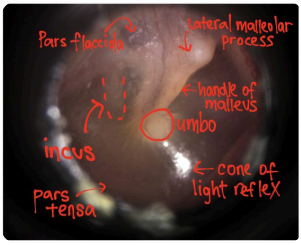
A

- **Umbo**

Why: The umbo is the "belly button" of the TM, at the tip of the malleus handle.

Question 3 (slide 3)

SINGLE ANSWER



In the LEFT ear, where is the normal cone of light?

A. It is absent in a normal ear

B. About 7 o'clock

C. About 12 o'clock

D. About 5 o'clock

Swipe for answer →

Answer 3

B

- **About 7 o'clock**

Why: "Right 5, Left 7." The cone of light points anterior-inferior.

Slide 4: what is this?



Name it before you swipe →

Normal left ear (incus visible)



Photo: Hawke, M. Hawke Library (course slides).

What it is: Another normal eardrum. It's so clear that you can see the incus (a middle ear bone) behind it.

What you see:

- Translucent drum; the incus shows through behind it
- A whiter, more opaque area around the umbo, which is NORMAL (there's more cartilage there)
- No fluid, no bulging

Normal left ear (incus visible)

> 🧠 **"See-through = Safe." If you can see the bones BEHIND the drum, there's no fluid in the way.**

Exam tips:

- Don't call a little whiteness around the umbo abnormal. That's normal anatomy.
- Compare: in OME/AOM, fluid or pus blocks your view of the middle ear.

Question 4 (slide 4)

SINGLE ANSWER



During otoscopy the nurse can see the incus through a translucent TM, and the area around the umbo is slightly whiter. The best interpretation is:

A. Acute otitis media

B. Tympanosclerosis

C. Normal tympanic membrane

D. Otitis media with effusion

Swipe for answer →

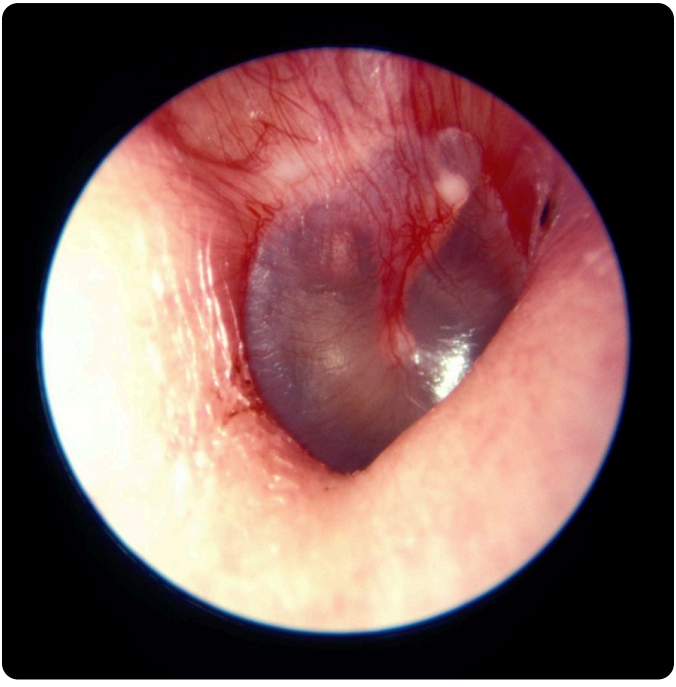
Answer 4

C

- **Normal tympanic membrane**

Why: Seeing middle ear structures through a translucent drum means there's no fluid. Slight opacity at the umbo is normal cartilage.

Slide 5: what is this?



Name it before you swipe →

Slide 5

Red TM from crying or fever (NOT AOM)

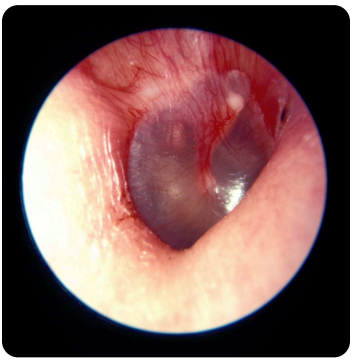


Photo: Hawke, M. Hawke Library (course slides).

What it is: The right ear of a child who is crying or has a fever. The drum looks RED from blood vessels, but this is NOT an ear infection.

What you see:

- Red, dilated blood vessels coming down from the top of the ear canal
- The drum is NOT bulging
- Landmarks are still visible
- No pus or fluid level

Slide 5: remember it

Red TM from crying or fever (NOT AOM)

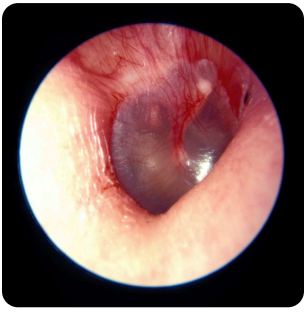
> 🧠 **"Red alone = Rubbish." Crying, fever, and rubbing make the TM red. You need a BULGE (or new drainage) to call it AOM.**

Exam tips:

- Don't give antibiotics for redness alone.
- The study guide: "Redness alone is insufficient. Crying/fever can redden a TM."

Question 5 (slide 5)

SINGLE ANSWER



A screaming 18-month-old has a red TM with visible landmarks and no bulging. What is the best interpretation?

A. Tympanic membrane perforation

B. Otitis externa

C. Acute otitis media; start amoxicillin

D. Redness from crying; this alone is not AOM

Swipe for answer →

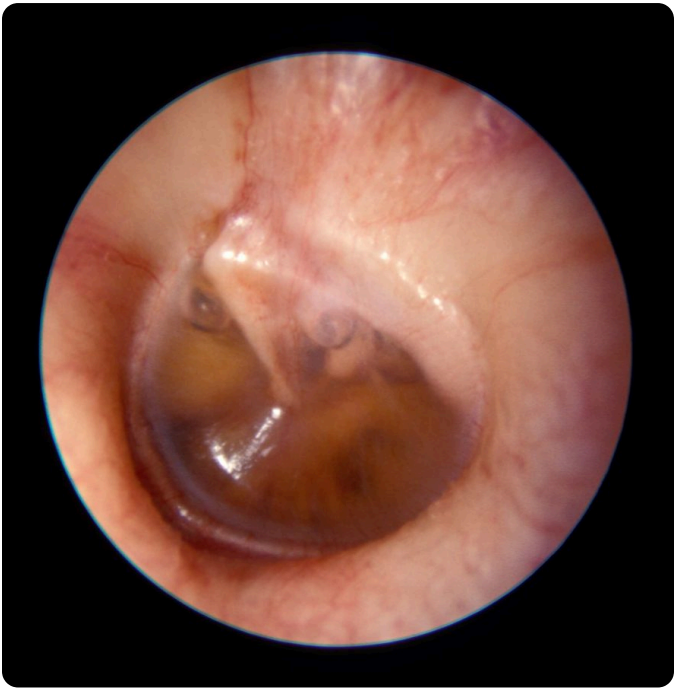
Answer 5

D

- **Redness from crying; this alone is not AOM**

Why: "Red alone = Rubbish." AOM requires bulging (or new otorrhea, or mild bulging with ear pain or marked erythema).

Slide 6: what is this?



Name it before you swipe →

Slide 6

Air-fluid level with bubbles (OME)



Photo: Hawke, M. Hawke Library (course slides).

What it is: Left ear with fluid behind the eardrum and air bubbles: one in front of and one behind the malleus. This is otitis media WITH EFFUSION (OME): fluid without infection.

What you see:

- Air bubbles behind the drum
- A line where the air meets the fluid (air-fluid level)
- The TM is NOT bulging and the cone of light is still there
- No signs of acute illness

Air-fluid level with bubbles (OME)

> 🧠 **"Bubbles = fluid without Fire." Bubbles or a fluid line + a calm child = OME, not AOM.**

Exam tips:

- OME: watch for about 3 months; check hearing if it persists. No routine antibiotics, decongestants, antihistamines, or steroids.
- OME can cause CONDUCTIVE hearing loss and speech delay if it lasts.

Question 6 (slide 6)

SINGLE ANSWER



A well 3-year-old has air bubbles behind a non-bulging TM. No fever or ear pain. What does the nurse anticipate?

A. Amoxicillin 80–90 mg/kg/day

B. Watchful waiting, with a hearing check if it persists

C. Oral decongestants and antihistamines

D. Amoxicillin-clavulanate

Swipe for answer →

Answer 6

B

- **Watchful waiting, with a hearing check if it persists**

Why: OME = fluid without fire. Observe about 3 months. Medications are not routinely used.

Slide 7: what is this?



Name it before you swipe →

Slide 7

Amber fluid with air-fluid level (OME)



Photo: Hawke, M. Hawke Library (course slides).

What it is: Right ear with amber (yellow-brown) fluid filling the lower part of the middle ear, with an air-fluid level. Also OME.

What you see:

- Amber/yellow fluid behind the drum, mostly at the bottom
- A visible fluid line
- No bulging; landmarks are still seen

Amber fluid with air-fluid level (OME)

> 🧠 **"Amber, not Angry." Amber fluid with a calm child = OME. An angry, bulging drum = AOM.**

Exam tips:

- Type of hearing loss: CONDUCTIVE (sound can't pass through the fluid).
- Fluid left over after AOM treatment does NOT automatically need more antibiotics.

Question 7 (slide 7)

SINGLE ANSWER



Amber fluid with an air-fluid level behind the TM most likely causes which type of hearing loss?

A. Conductive

B. Sensorineural

C. None; fluid doesn't affect hearing

D. Mixed

Swipe for answer →

Answer 7

A

- **Conductive**

Why: Conductive = Cerumen and fluid (outer/middle ear). Sensorineural = Snail (cochlea) and nerve.

Slide 8: what is this?



Name it before you swipe →

Acute otitis media: moderate bulging



Photo: Hawke, M. Hawke Library (course slides).

What it is: Left ear with a moderately BULGING eardrum and pus (exudate) behind it. This is ACUTE OTITIS MEDIA.

What you see:

- The TM bulges outward toward you
- Cloudy/opaque, with pus behind it
- The umbo looks pulled in (central dimple) and the landmarks are LOST
- Red and inflamed

Acute otitis media: moderate bulging

> 🧠 **"Bulge = AOM." Think of an inflated balloon: Bulging, Opaque, Lost landmarks, Doesn't move ("BOLD").**

Exam tips:

- Bugs: "Stop Hurting Me": S. pneumoniae, H. influenzae (nontypeable), Moraxella.
- First line: amoxicillin 80–90 mg/kg/day divided BID. ALWAYS treat the pain.

Question 8 (slide 8)

SELECT ALL THAT APPLY



Which otoscopic findings support a diagnosis of acute otitis media?

A. Reduced mobility on pneumatic otoscopy

B. Opaque TM with pus behind it

C. Loss of bony landmarks

D. Moderate bulging of the TM

E. Red TM in a crying child with normal landmarks

F. Pearly gray, translucent TM

Swipe for answer →

Answer 8

A, B, C, D

- **Reduced mobility on pneumatic otoscopy**
- **Opaque TM with pus behind it**
- **Loss of bony landmarks**
- **Moderate bulging of the TM**

Why: Bulge = AOM (BOLD: Bulging, Opaque, Lost landmarks, Doesn't move). Redness alone is not AOM.

Question 9 (slide 8)

SINGLE ANSWER



A 14-month-old has this picture in BOTH ears, with fever of 38.4°C. What is the plan?

A. Refer for ear tubes

B. Ear drops only

C. Observe for 48–72 hours

D. Antibiotics plus pain control

Swipe for answer →

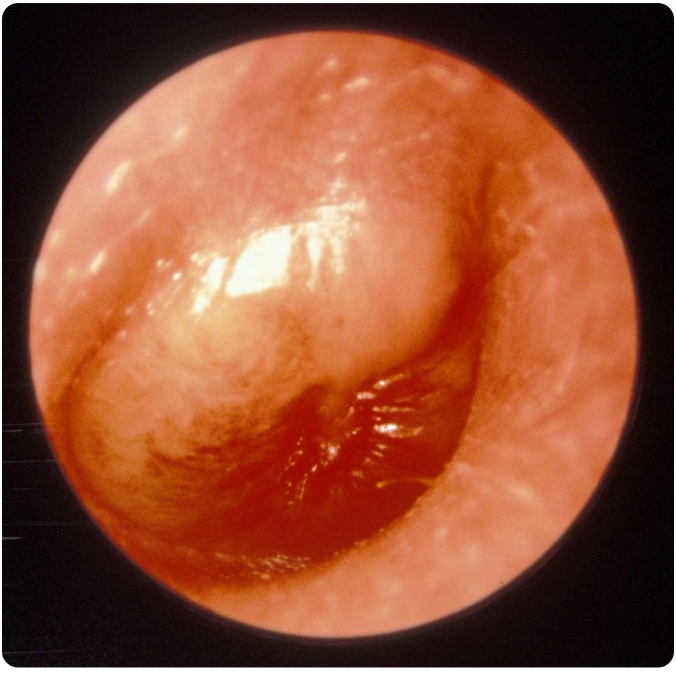
Answer 9

D

- **Antibiotics plus pain control**

Why: Under 2 years with AOM in both ears → antibiotics, even if nonsevere. Always treat the pain.

Slide 9: what is this?



Name it before you swipe →

Slide 9

AOM with bulging and TM hemorrhage

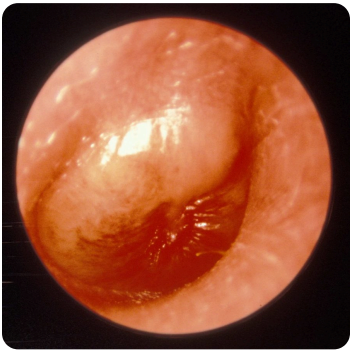


Photo: Hawke, M. Hawke Library (course slides).

What it is: Right ear with a BULGING eardrum, lost landmarks, and bleeding INTO the eardrum (hemorrhage). Still acute otitis media, a very inflamed one.

What you see:

- Bulging, red-orange TM
- No landmarks visible
- Dark red streaks/blotches of blood within the drum

AOM with bulging and TM hemorrhage

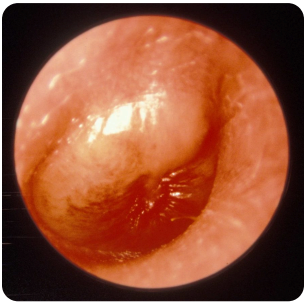
> 🧠 **"Bulge + Blood = Bad AOM." Bleeding in the drum doesn't change the diagnosis. It's AOM.**

Exam tips:

- Expect a lot of pain. Give acetaminophen or ibuprofen (ibuprofen only if 6 months or older).
- Severe AOM (fever of 39°C or higher, or severe pain, or pain for 48 hours or more) → antibiotics at any age.

Question 10 (slide 9)

SINGLE ANSWER



A 3-year-old has a bulging TM with hemorrhage, severe ear pain, and a temperature of 39.4°C. Which is correct?

A. Observation is preferred because the child is over 2

B. No pain medication until the diagnosis is confirmed

C. Antibiotics are indicated because the illness is severe

D. This is otitis externa

Swipe for answer →

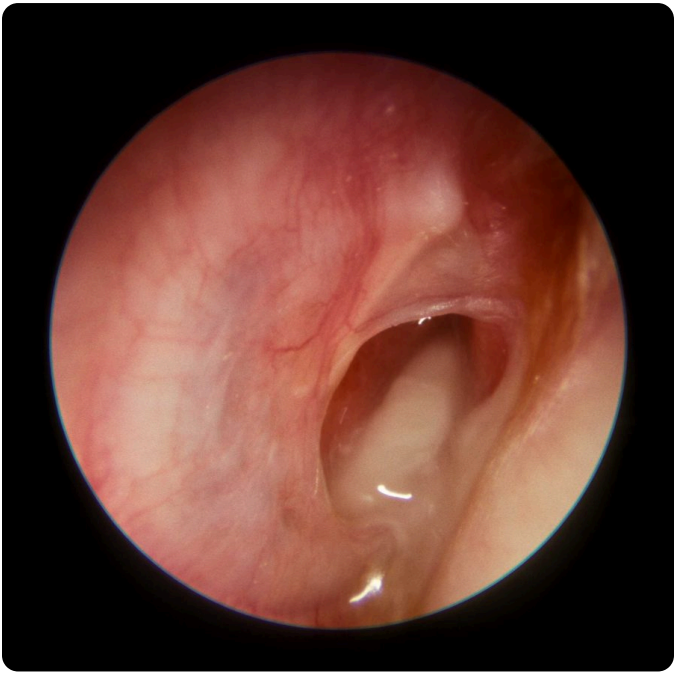
Answer 10

C

- **Antibiotics are indicated because the illness is severe**

Why: Severe AOM = "39° or 48 hrs" (or otorrhea) → antibiotics at any age. Always treat pain.

Slide 10: what is this?



Name it before you swipe →

Perforated tympanic membrane



Photo: Hawke, M. Hawke Library (course slides).

What it is: A right eardrum with a HOLE (perforation). The lateral process and malleus are still visible. In kids this often happens when pressure from AOM bursts the drum.

What you see:

- A dark opening in the drum
- Landmarks (malleus) still seen
- There may be drainage (otorrhea) in the canal

Perforated tympanic membrane

> 🧠 **"Pop → Pus → Pain relief." The drum pops, pus drains, and the pain suddenly gets better. Then think: "No water in a hole": no irrigation, no swimming without protection.**

Exam tips:

- New drainage (otorrhea) not from otitis externa supports AOM → antibiotics are indicated.
- Do NOT irrigate an ear with a suspected perforation (one of the 3 B's: Battery, Beans, Broken eardrum).
- Most small perforations heal on their own; follow up and check hearing.

Question 11 (slide 10)

SINGLE ANSWER



A child with AOM suddenly has less ear pain, and there is yellow drainage in the ear canal. What does the nurse suspect?

A. Cerumen impaction

B. Otitis media with effusion

C. The infection has resolved

D. Tympanic membrane perforation

Swipe for answer →

Answer 11

D

- **Tympanic membrane perforation**

Why: "Pop → Pus → Pain relief." A perforation releases pressure and causes drainage.

Question 12 (slide 10)

SELECT ALL THAT APPLY



Which actions are appropriate for a child with a perforated TM?

A. Use cotton swabs to clean deep in the canal

B. Irrigate to remove the drainage

C. Keep water out of the ear

D. Arrange follow-up of hearing and healing

E. Avoid ear irrigation

Swipe for answer →

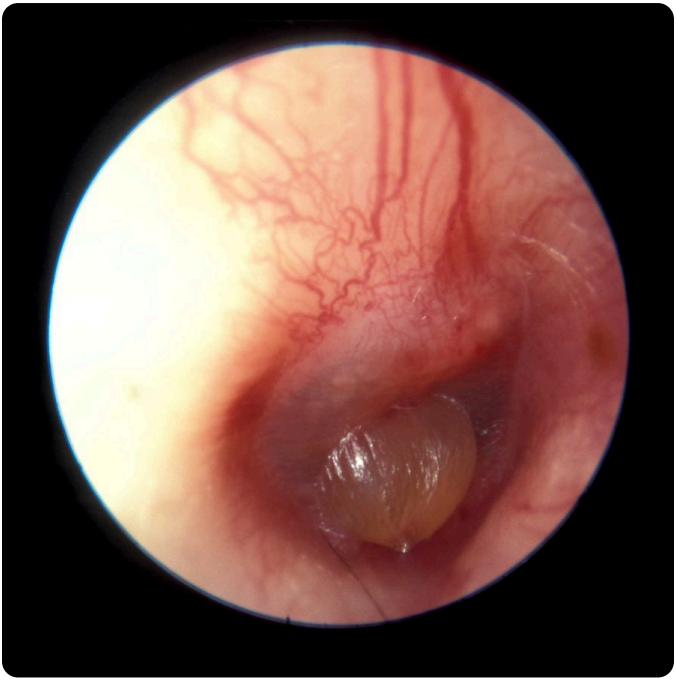
Answer 12

C, D, E

- **Keep water out of the ear**
- **Arrange follow-up of hearing and healing**
- **Avoid ear irrigation**

Why: Never irrigate a Broken eardrum. Keep it dry, nothing in the canal, and follow up.

Slide 11: what is this?



Name it before you swipe →

Bullous myringitis

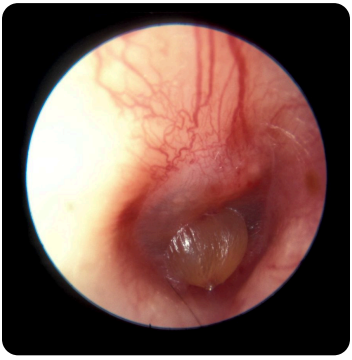


Photo: Hawke, M. Hawke Library (course slides).

What it is: A fluid-filled blister (bulla) ON the eardrum. This is bullous myringitis, an inflamed drum that usually goes along with acute otitis media.

What you see:

- A round, shiny, fluid-filled bubble sitting ON the drum (not behind it)
- Red, inflamed vessels across the drum

Bullous myringitis

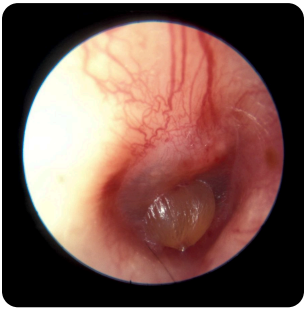
> 🧠 **"Bulla on the drum = Big pain." Bubbles ON the drum = bullous myringitis. Bubbles BEHIND the drum = OME.**

Exam tips:

- VERY painful, so pain control is the priority.
- It is caused by the same bacteria as AOM and is usually treated the same way as AOM.

Question 13 (slide 11)

SINGLE ANSWER



The nurse sees a fluid-filled blister ON the surface of a red TM in a child with severe ear pain. This is:

A. Cholesteatoma

B. Otitis media with effusion

C. Bullous myringitis

D. Tympanosclerosis

Swipe for answer →

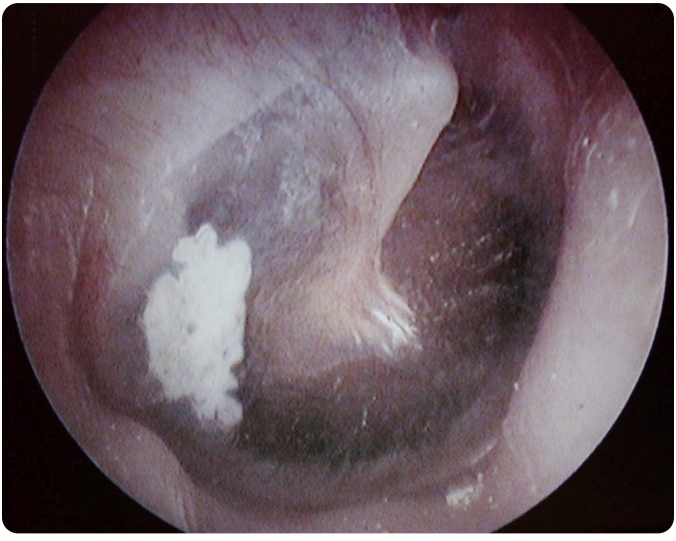
Answer 13

C

- **Bullous myringitis**

Why: Bulla ON the drum = bullous myringitis (very painful, treated like AOM). Bubbles BEHIND the drum = OME.

Slide 12: what is this?



Name it before you swipe →

Tympanosclerosis

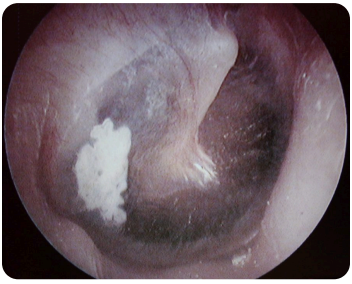


Photo: Descouens, D. Tympanosclerose, CC BY-SA 3.0, Wikimedia Commons.

What it is: A white, chalky, irregular patch on the back part of the eardrum. This is tympanosclerosis: calcium scarring left over from old ear infections or ear tubes.

What you see:

- A bright WHITE, chalk-like, irregular oval patch
- The rest of the drum is normal, with normal bony landmarks
- NO fluid behind the drum

Tympanosclerosis

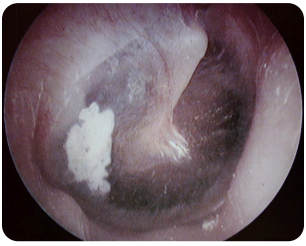
> 🧠 **"Chalk on the chalkboard = an old Scar." White patch + normal everything else = tympanosclerosis. It's history, not an infection.**

Exam tips:

- Usually harmless and needs no treatment. Large plaques can occasionally affect hearing.
- Common after repeated AOM or tympanostomy tubes. Do not give antibiotics for it.

Question 14 (slide 12)

SINGLE ANSWER



A child with a history of ear tubes has a chalky white plaque on the TM, normal landmarks, and no fluid. The nurse should:

A. Start amoxicillin

B. Irrigate the ear

C. Recognize tympanosclerosis, which usually needs no treatment

D. Refer immediately for cholesteatoma surgery

Swipe for answer →

Answer 14

C

- **Recognize tympanosclerosis, which usually needs no treatment**

Why: "Chalk on the chalkboard = an old Scar." Tympanosclerosis comes from previous infection or tubes and is usually harmless.

Slide 13: what is this?



Name it before you swipe →

Attic cholesteatoma



Photo: Hawke, M. Attic Cholesteatoma, CC BY 4.0, Wikimedia Commons.

What it is: Left ear with a cherry-red lump of granulation tissue at the top of the eardrum (pars flaccida). Behind it is a perforation leading to a CHOLESTEATOMA: a growing pocket of trapped skin in the "attic" of the middle ear. Ongoing drainage has made the drum look thick and white.

What you see:

- A small cherry-red mass at the top of the drum (pars flaccida)
- White, thickened, swollen drum surface from chronic discharge
- Chronic drainage

Attic cholesteatoma

> 🧠 **Cholesteatoma = "Skin cyst in the attic" with the 3 C's: Chronic smelly drainage, Can't hear (hearing loss), Cut it out (ENT surgery).**

Exam tips:

- It is NOT cancer, but it slowly destroys bone and can damage hearing, the facial nerve, or spread infection.
- Needs ENT referral. Antibiotic drops alone won't cure it. The study guide lists "suspected cholesteatoma" as needing further assessment.

Question 15 (slide 13)

SELECT ALL THAT APPLY



Which findings should make the nurse suspect a cholesteatoma?

A. Pearly gray, mobile TM

B. Hearing loss

C. Mass or perforation in the pars flaccida

D. A single episode of ear pain that resolved

E. Chronic, foul-smelling ear drainage

Swipe for answer →

Answer 15

B, C, E

- **Hearing loss**
- **Mass or perforation in the pars flaccida**
- **Chronic, foul-smelling ear drainage**

Why: 3 C's: Chronic smelly drainage, Can't hear, Cut it out. It often starts in the pars flaccida (attic).

Question 16 (slide 13)

SINGLE ANSWER



What is the expected management of a cholesteatoma?

A. ENT referral for surgical removal

B. Observation for 3 months

C. Ear irrigation

D. Amoxicillin for 10 days

Swipe for answer →

Answer 16

A

- **ENT referral for surgical removal**

Why: "Cut it out": it keeps growing and destroying bone, so it needs ENT and usually surgery.

Slide 14: what is this?



Name it before you swipe →

Slide 14

Acute otitis externa with pinna cellulitis/perichondritis



Course slide (Appendix B3).

What it is: An outer ear infection (otitis externa) that has spread: the whole outer ear (pinna) is red and swollen, with crusted drainage coming out of the canal. Spread to the ear cartilage = perichondritis.

What you see:

- Red, swollen, shiny pinna
- Yellow crusting/drainage from the ear canal
- The problem is the OUTER ear, not the eardrum

Acute otitis externa with pinna cellulitis/perichondritis

> 🧠 **"Pull the ear, it hurts." When the WHOLE ear is red and puffy, the infection has "escaped the canal" and needs more than drops.**

Exam tips:

- Spreading cellulitis/perichondritis needs prompt evaluation and often systemic antibiotics, since cartilage damage can deform the ear.
- Compare with mastoiditis: swelling BEHIND the ear plus the ear pushed forward and out.

Question 17 (slide 14)

SINGLE ANSWER



A child has a red, swollen, tender outer ear with crusted drainage from the canal. Pulling on the ear causes severe pain. This is most consistent with:

A. Otitis externa with spreading cellulitis

B. Acute otitis media

C. Otitis media with effusion

D. Tympanosclerosis

Swipe for answer →

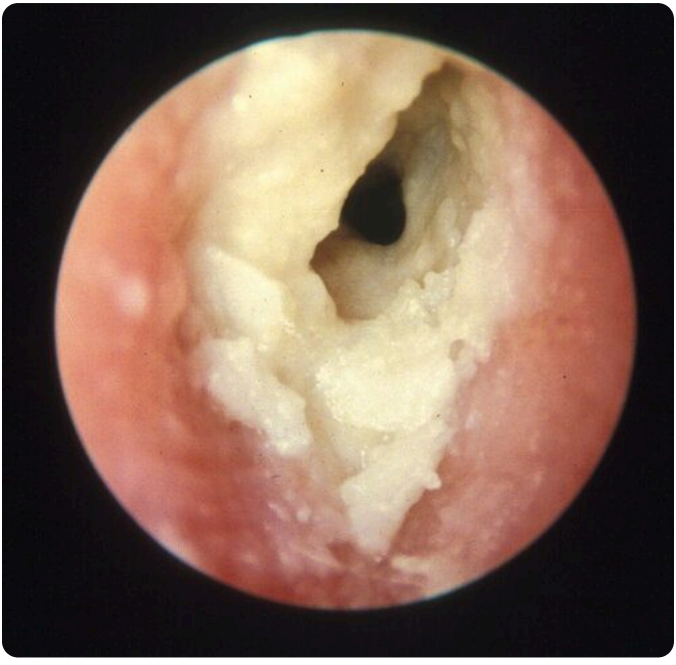
Answer 17

A

- **Otitis externa with spreading cellulitis**

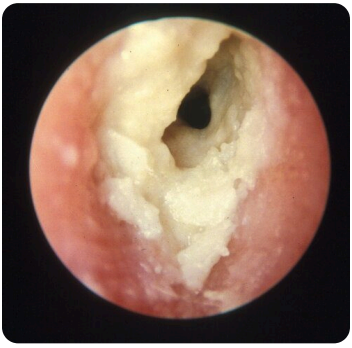
Why: Pain when moving the pinna or pressing the tragus = otitis externa. A red, swollen pinna means it's spreading (cellulitis/perichondritis).

Slide 15: what is this?



Name it before you swipe →

Bacterial acute otitis externa (inside the canal)



Course slide (Appendix B3).

What it is: A view inside the ear canal in bacterial otitis externa ("swimmer's ear"). The canal is red and swollen and filled with creamy, custard-like discharge.

What you see:

- Thick, creamy white-yellow "custard" debris in the canal
- Red, swollen canal walls
- The eardrum may be hard to see

Bacterial acute otitis externa (inside the canal)

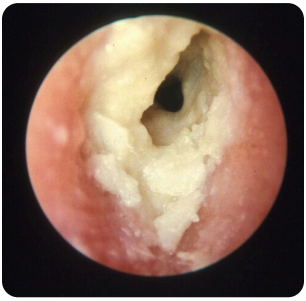
> 🧠 **"Swimmer's ear = Custard in the Canal." Treat with "Drops, Dry, Don't poke": topical antibiotic drops, keep the ear dry, no cotton swabs.**

Exam tips:

- Treatment: topical otic drops; keep the ear dry; avoid putting anything in the ear. A wick may be needed if the canal is very swollen.
- Before choosing drops, check whether the TM is intact.

Question 18 (slide 15)

SELECT ALL THAT APPLY



Which teaching is appropriate for a child with otitis externa?

A. Keep the ear dry

B. Swim daily to rinse the canal

C. Avoid cotton swabs in the canal

D. Use the prescribed ear drops

E. Irrigate the ear at home

Swipe for answer →

Answer 18

A, C, D

- **Keep the ear dry**
- **Avoid cotton swabs in the canal**
- **Use the prescribed ear drops**

Why: "Drops, Dry, Don't poke." Water and instruments make otitis externa worse.

Slide 16: what is this?



Name it before you swipe →

Otitis externa (swollen canal opening)



Course slide (Appendix B3).

What it is: A child's outer ear where the canal opening is red, swollen, and narrowed, with debris. This is otitis externa.

What you see:

- Red, swollen entrance to the ear canal
- Debris/discharge at the opening
- The pinna itself looks mostly normal

Otitis externa (swollen canal opening)

> 🧠 **"Tug test": tug the ear or push the tragus. If it HURTS → otitis EXTERNA (outside). AOM usually doesn't hurt when you move the ear.**

Exam tips:

- The fastest way to tell OE from AOM is pain with tragus pressure or pinna movement.
- Risk factors: swimming, moisture, scratching, cotton swabs.

Question 19 (slide 16)

SINGLE ANSWER



Which assessment finding best distinguishes otitis externa from acute otitis media?

A. Recent cold

B. Pain when the tragus is pressed or the pinna is moved

C. Irritability

D. Fever

Swipe for answer →

Answer 19

B

- **Pain when the tragus is pressed or the pinna is moved**

Why: "Pull the ear, it hurts" = otitis externa. AOM doesn't usually hurt with ear movement.

Slide 17: what is this?



Name it before you swipe →

Slide 17

Otitis externa (narrowed canal on otoscopy)



Course slide (Appendix B3).

What it is: An otoscope view down a red, swollen ear canal that has narrowed to a small opening. The eardrum can't be seen well. This is otitis externa.

What you see:

- Canal walls red and swollen
- The canal has narrowed to a small slit
- The eardrum is hard or impossible to see

Otitis externa (narrowed canal on otoscopy)

> 🧠 **"Can't see the drum? Check the Canal." A swollen canal blocks your view. That's an outside-the-drum problem.**

Exam tips:

- A very swollen canal may need a wick so the drops can get in.
- Insert the otoscope gently, because the canal is extremely tender.

Question 20 (slide 17)

SINGLE ANSWER



On otoscopy, the canal is red and swollen shut and the TM cannot be visualized. The child cries when the ear is touched. What does the nurse anticipate?

A. Ear irrigation

B. Topical otic drops, possibly with a wick

C. Oral amoxicillin for AOM

D. Watchful waiting for 3 months

Swipe for answer →

Answer 20

B

- **Topical otic drops, possibly with a wick**

Why: Otitis externa = topical therapy. A wick helps drops get into a swollen canal. Don't irrigate.